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Formation method of shallow trench isolation

Abstract

A method of forming a semiconductor device includes etching trenches in a substrate to form fin structures, depositing a liner layer to line the trenches, filling the trenches with an insulating layer, performing an ion implantation process to the insulating layer, after performing the ion implantation process, recessing the insulating layer to form shallow trench isolation (STI) regions adjacent the fin structures, and forming a gate crossing the fin structures.

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Background/Summary

BACKGROUND

(1) Semiconductor devices are used in a variety of electronic applications, such as, for example, personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon.

(2) The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates an example of gate-all-around field-effect transistors (GAA-FETs) in a three-dimensional view, in accordance with some embodiments.

(3) FIGS. 2, 3, 13, 14A, 21A, 22A, 23A, 24A and 25A are cross-sectional views at intermediate fabrication stages, illustrating reference cross-section A-A' illustrated in FIG. 1 that extends through a gate region along a longitudinal axis of the gate region.

(4) FIGS. 4, 5, 6A, 7A and 7B are three-dimensional views at intermediate fabrication stages, in accordance with some embodiments.

(5) FIG. 6B is a graph illustrating an impurity concentration in the insulating layer as a function of a distance from a top of the insulating layer.

(6) FIG. 8 is a bar chart illustrating etch rates over the liner layer and the non-ion implanted/ion-implanted insulating layers in various embodiments constructed according to various aspects of the present disclosure.

(7) FIGS. 9-11 illustrate diagrams of penetration depths versus energy performed during the ion implantation.

(8) FIG. 12 shows a diagram of depth R3 versus the etch rate difference between the liner layer and the insulating layer in accordance with some embodiments.

(9) FIGS. 14B, 15B, 16B, 17B, 18B, 19B, 20B, 21B, 22B, 23B, 23C, 24B and 25B are cross-sectional views at intermediate fabrication stages, illustrating reference cross-section B-B' illustrated in FIG. 1 that extends through a fin along a longitudinal axis of the fin.

(10) FIGS. 15A, 16A, 17A, 18A, 19A, 20A, 20C, 21C are cross-sectional views at intermediate fabrication stages, illustrating reference cross-section C-C' illustrated in FIG. 1 that extends through source/drain regions along the longitudinal direction of the gate region.

DETAILED DESCRIPTION

(11) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and

second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(12) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 230 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. As used herein, “around,” “about,” “approximately,” or “substantially” may generally mean within 20 percent, or within 10 percent, or within 5 percent of a given value or range. Numerical quantities given herein are approximate, meaning that the term “around,” “about,” “approximately,” or “substantially” can be inferred if not expressly stated.

(13) The present disclosure is generally related to integrated circuit (IC) structures and methods of forming the same, and more particularly to fabricating gate-all-around (GAA) transistors having different high-k gate dielectric compositions. It is also noted that the present disclosure presents embodiments in the form of multi-gate transistors. Multi-gate transistors include those transistors whose gate structures are formed on at least two-sides of a channel region. These multi-gate devices may include a p-type metal-oxide-semiconductor device or an n-type metal-oxide-semiconductor device. Specific examples may be presented and referred to herein as fin field effect transistors (FinFET), on account of their fin-like structure. Also presented herein are embodiments of a type of multi-gate transistor referred to as a gate-all-around (GAA) device. A GAA device includes any device that has its gate structure, or portion thereof, formed on 4-sides of a channel region (e.g., surrounding a portion of a channel region). Devices presented herein also include embodiments that have channel regions disposed in nanosheet channel(s), nanowire channel(s), and/or other suitable channel configuration. Presented herein are embodiments of devices that may have one or more channel regions (e.g., nanosheets) associated with a single, contiguous gate structure. However, one of ordinary skill would recognize that the teaching can apply to a single channel (e.g., single nanosheet) or any number of channels. One of ordinary skill may recognize other examples of semiconductor devices that may benefit from aspects of the present disclosure.

(14) The gate all around (GAA) transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

(15) As scales of the fin width in fin field effect transistors (FinFET) decreases, channel width variations might cause mobility loss. GAA transistors, such as nanosheet transistors are being studied as an alternative to fin field effect transistors. In a nanosheet transistor, the gate of the transistor is made all around the channel (e.g., a nanosheet channel or a nanowire channel) such that the channel is surrounded or encapsulated by the gate. Such a transistor has the advantage of improving the electrostatic control of the channel by the gate, which also mitigates leakage currents.

(16) In transistors, shallow trench isolations (STIs) are used to separate and isolate active areas on a semiconductor wafer from each other. STIs may be formed by etching trenches, overfilling the trenches with a dielectric layer such as oxide, and then removing any excess dielectric with a

process such as chemical mechanical polishing (CMP) or etching in order to remove the dielectric outside trenches. A liner layer (e.g., oxide) is used to line the trenches for the STIs to protect the underlying substrate from subsequent process. For example, the liner layer may be made of a dense film using physical vapor deposition to shield the underlying substrate from oxidation chemistry in subsequent processing. By contrast, the STIs may be formed by a flowable chemical vapor deposition process to enhance a gap fill capability. The SIT oxide formed from flowable deposition has a faster etch rate than the liner oxide formed from PVD. Such etch rate difference leads to “dishing effect” of the STIs in an STI etch back process, which means the resultant STIs may have a smaller thickness at central regions of STIs than at peripheral regions of STIs.

(17) Therefore, the present disclosure in various embodiments reduces an etch rate of the STIs by using an ion implantation process. For example, the ions can introduce additional atoms and breaking Si—O bonding in the STIs, both of which result in an enhanced etch resistance for the STIs.

(18) FIG. 1 illustrates an example of GAA-FETs (e.g., nanowire FETs, nanosheet FETs, or the like) in a three-dimensional view, in accordance with some embodiments. The GAA-FETs comprise nanostructures **104** (e.g., nanosheets, nanowires, nanorings, nanoslabs, or other structures having nano-scale size (e.g., a few nanometers)) over fins **102** on a substrate **100** (e.g., a semiconductor substrate), wherein the nanostructures **104** act as channel regions for the GAA-FETs. The nanostructure **104** may include p-type nanostructures, n-type nanostructures, or a combination thereof. Isolation regions **106** are disposed between adjacent fins **102**, which may protrude above and from between neighboring isolation regions **106**. A liner layer **105** is formed between the isolation regions **106** and the fins **102**. Although the isolation regions **106** are described/illustrated as being separate from the substrate **100**, as used herein, the term “substrate” may refer to the semiconductor substrate alone or a combination of the semiconductor substrate and the isolation regions. Additionally, although a bottom portion of the fins **102** are illustrated as being single, continuous materials with the substrate **100**, the bottom portion of the fins **102** and/or the substrate **100** may comprise a single material or a plurality of materials. In this context, the fins **102** refer to the portion extending between the neighboring isolation regions **106**.

(19) Gate dielectrics **110** are over top surfaces of the fins **102** and along top surfaces, sidewalls, and bottom surfaces of the nanostructures **104**. Gate electrodes **112** are over the gate dielectrics **110**. Epitaxial source/drain regions **108** are disposed on the fins **102** on opposing sides of the gate dielectric **110** and the gate electrodes **112**.

(20) FIG. 1 further illustrates reference cross-sections that are used in later figures. Cross-section A-A' is along a longitudinal axis of a gate electrode **112** and in a direction, for example, perpendicular to the direction of current flow between the epitaxial source/drain regions **108** of a GAA-FET. Cross-section B-B' is perpendicular to cross-section A-A' and is parallel to a longitudinal axis of a fin **102** of the GAA-FET and in a direction of, for example, a current flow between the epitaxial source/drain regions **108** of the GAA-FET. Cross-section C-C' is parallel to cross-section A-A' and extends through epitaxial source/drain regions of the GAA-FETs. Subsequent figures refer to these reference cross-sections for clarity.

(21) Some embodiments discussed herein are discussed in the context of GAA-FETs formed using a gate-last process. In other embodiments, a gate-first process may be used. Also, some embodiments contemplate aspects used in planar devices, such as planar FETs or in fin field-effect transistors (FinFETs).

(22) FIGS. 2, 3, 13, 14A, 21A, 22A, 23A, 24A and 25A are cross-sectional views at intermediate fabrication stages, illustrating reference cross-section A-A' illustrated in FIG. 1 that extends through a gate region along a longitudinal axis of the gate region. FIGS. 4-7B are three-dimensional views at intermediate fabrication stages, in accordance with some embodiments. FIGS. 14B, 15B, 16B, 17B, 18B, 19B, 20B, 21B, 22B, 23B, 23C, 24B and 25B are cross-sectional views at intermediate fabrication stages, illustrating reference cross-section B-B' illustrated in FIG. 1 that

extends through a fin along a longitudinal axis of the fin. FIGS. 15A, 16A, 17A, 18A, 19A, 20A, 20C, 21C are cross-sectional views at intermediate fabrication stages, illustrating reference cross-section C-C' illustrated in FIG. 1 that extends through source/drain regions along the longitudinal direction of the gate region.

(23) In FIG. 2, a substrate **100** is provided. The substrate **100** may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The substrate **100** may be a wafer, such as a silicon wafer. Generally, an SOI substrate is a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **100** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; or combinations thereof.

(24) The substrate **100** has a first device region **1001** and a second region **1002**. The first device region **1001** is a region in which first transistors will reside, and the second device region **1002** is a region in which second transistors will reside. In some embodiments, the first transistors are different from the second transistors at least in threshold voltage. For example, first transistors in the first device region **1001** may be High Voltage (HV) devices (e.g., I/O devices), and second transistors in the second device region **1002** may be Low Voltage (LV) devices (e.g., logic devices). In some other embodiments, the first transistors are different from the second transistors at least in conductivity type. For example, the first device region **1001** can be for forming n-type devices, such as n-channel metal-oxide-semiconductor (NMOS) transistors, e.g., n-type GAA-FETs, and the second device region **1002** can be for forming p-type devices, such as p-channel metal-oxide-semiconductor (PMOS) transistors, e.g., p-type GAA-FETs.

(25) The first device region **1001** may be separated from the second device region **1002**, and any number of device features (e.g., other active devices, doped regions, isolation structures, etc.) may be disposed between the first device region **1001** and the second device region **1002**. Although one first device region **1001** and one second device region **1002** are illustrated, any number of first device regions **1001** and second device regions **1002** may be provided.

(26) Further in FIG. 2, a multi-layer stack **201** is formed over the substrate **100**. The multi-layer stack **201** includes alternating layers of first semiconductor layers **202A-D** (collectively referred to as first semiconductor layers **202**) and second semiconductor layers **204A-D** (collectively referred to as second semiconductor layers **204**). For purposes of illustration and as discussed in greater detail below, the first semiconductor layers **202** will be removed and the second semiconductor layers **204** will be patterned to form channel regions of GAA-FETs.

(27) The first semiconductor layers **202** and the second semiconductor layers **204** may include one or more layers of Si, Ge, SiGe, GaAs, InSb, GaP, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb or InP. In some embodiments, the first semiconductor layers **110** and the second semiconductor layers **112** are made of Si, a Si compound, SiGe, Ge or a Ge compound. The first semiconductor layers **202** include a first composition and the second semiconductor layers **204** include a second composition different from the first composition. The first and second compositions have different oxidation rates and/or etch selectivity. For example, the first semiconductor layers **202** may include SiGe and the second semiconductor layers **204** may include Si.

(28) The multi-layer stack **201** is illustrated as including four layers of each of the first semiconductor layers **202** and the second semiconductor layers **204** for illustrative purposes. In some embodiments, the multi-layer stack **201** may include any number of the first semiconductor

layers **202** and the second semiconductor layers **204**. Each of the layers of the multi-layer stack **201** may be epitaxially grown using a process such as chemical vapor deposition (CVD), atomic layer deposition (ALD), vapor phase epitaxy (VPE), molecular beam epitaxy (MBE), or the like. In various embodiments, the second semiconductor layers **204** may be formed of a semiconductor material suitable for serving as channel regions of GAA-FETs, such as silicon, silicon carbon, silicon germanium, or the like.

(29) The first semiconductor materials and the second semiconductor materials may be materials having a high-etch selectivity to one another. As such, the first semiconductor layers **202** of the first semiconductor material may be removed without significantly removing the second semiconductor layers **204** of the second semiconductor material, thereby allowing the second semiconductor layers **204** to serve as channel regions of GAA-FETs.

(30) The topmost second semiconductor layer **204D** can function as a dummy semiconductor layer or sacrificial layer to protect the channel regions of GAA-FETs from being damaged during a subsequent ion implantation process and has a thickness different from other second semiconductor layers **204A-204C**. For example, the topmost second semiconductor layer **204D** has a thickness t_1 greater than the thicknesses t_2 of the second semiconductor layers **204A-204C**. In some embodiments, the topmost first semiconductor layer **202D** has the thickness t_1 in a range from 35 nm to 45 nm.

(31) A Pad layer **12** and a mask layer **14** may be formed on the multi-layer stack **201**. The pad layer **12** may be a thin film comprising silicon oxide formed, for example, using a thermal oxidation process. The pad layer **12** may act as an adhesion layer between multi-layer stack **201** and the mask layer **14**. The pad layer **12** may also act as an etch stop layer for etching the mask layer **14**. In an embodiment, the mask layer **14** is formed of silicon nitride, for example, using low-pressure chemical vapor deposition (LPCVD). In other embodiments, the mask layer **14** is formed by thermal nitridation of silicon, plasma enhanced chemical vapor deposition (PECVD), or plasma anodic nitridation. The mask layer **14** is used as a hard mask during subsequent photolithography processes. A photo resist (not shown) is formed on the mask layer **14** and is then patterned, forming openings in the photo resist. In some embodiments, the pad layer and the mask layer **14** collectively have a height in a range from 25 nm to 30 nm.

(32) Referring now to FIG. 3, fin structures **206** are formed in the substrate **100** and nanostructures **203** are formed in the multi-layer stack **201**, in accordance with some embodiments. In some embodiments, the nanostructures **203** and the fin structures **206** may be formed in the multi-layer stack **201** and the substrate **100**, respectively, by etching trenches **16** in the multi-layer stack **201** and the substrate **100** using the mask layer **14** and the pad layer **12** as an etch mask. Each fin structure **206** and overlying nanostructures **203** can be collectively referred to as a fin **207** extending from the substrate **100**. The trenches **16** separate neighboring nanostructures **203** and neighboring fin structures **206**. The trenches **16** have sidewalls **17** and a bottom surface **19**. The etching may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etching may be anisotropic. Forming the nanostructures **203** by etching the multi-layer stack **201** may further define first nanostructures **202A-D** (collectively referred to as the first nanostructures **202**) from the first semiconductor layers **202** and define second nanostructures **204A-D** (collectively referred to as the second nanostructures **204**) from the second semiconductor layers **204**. The first nanostructures **202** and the second nanostructures **204** may further be collectively referred to as nanostructures **203**.

(33) The fin structures **206** and the nanostructures **203** may be patterned by any suitable method. For example, the fin structures **206** and the nanostructures **203** may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one

embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fin structures **206**.

(34) FIG. 3 illustrates the fin structures **206** in the first device region **1001** and the second device region **1002** as having substantially equal widths for illustrative purposes. In some embodiments, widths of the fin structures **206** in the first device region **1001** may be greater or thinner than the fin structures **206** in the second device region **1002**. Further, while each of the fin structures **206** and the nanostructures **203** are illustrated as having a consistent width throughout, in other embodiments, the fin structures **206** and/or the nanostructures **203** may have tapered sidewalls such that a width of each of the fin structures **206** and/or the nanostructures **203** continuously increases in a direction towards the substrate **100**. In such embodiments, each of the nanostructures **203** may have a different width and be trapezoidal in shape.

(35) The process described above with respect to FIGS. 2 and 3 is just one example of how the fin structures **206** and the nanostructures **203** may be formed. In some embodiments, the fin structures **206** and/or the nanostructures **203** may be formed using a mask and an epitaxial growth process. For example, a dielectric layer can be formed over a top surface of the substrate **100**, and trenches can be etched through the dielectric layer to expose the underlying substrate **100**. Epitaxial structures can be epitaxially grown in the trenches, and the dielectric layer can be recessed such that the epitaxial structures protrude from the dielectric layer to form the fin structures **206** and/or the nanostructures **203**. The epitaxial structures may comprise the alternating semiconductor materials discussed above, such as the first semiconductor materials and the second semiconductor materials. In some embodiments where epitaxial structures are epitaxially grown, the epitaxially grown materials may be in situ doped during growth, which may obviate prior and/or subsequent implantations, although in situ and implantation doping may be used together.

(36) Additionally, the first semiconductor layers (and resulting nanostructures **202**) and the second semiconductor layers (and resulting nanostructures **204**) are illustrated and discussed herein as comprising the same materials in the second device region **1002** and the first device region **1001** for illustrative purposes only. As such, in some embodiments one or both of the first semiconductor layers and the second semiconductor layers may be different materials or formed in a different order in the first and second device regions **1001** and **1002**.

(37) A liner layer **18** and an insulating layer **20'** are deposited sequentially to line the trenches **16** in accordance with some embodiments. The liner layer is a conformal layer and is deposited on sidewalls **17** and the bottom surface **19** of the trenches **16**. After the deposition of the liner layer **18**, the insulating layer overfills the trench **16** and is over the mask layer **14**. A planarization process is performed to remove the insulating layer **20'** and the liner layer **18** outside the trenches **16**, as shown in FIG. 4. In some embodiments, the planarization process is a chemical mechanical polishing (CMP) process.

(38) The liner layer **18** and the insulating layer **20'** may be formed using different deposition methods. For example, the liner layer **18** may be deposited using physical vapor deposition (PVD) and may include silicon oxide, silicon nitride, silicon oxynitride or a combination thereof. The insulating layer **20'** may be deposited using flowable CVD (FCVD) and after the insulating layer **20'** is deposited, an anneal/curing step is performed, which converts the insulating layer **20'** from a flowable material into a solid material. Such difference in the formation methods between the liner layer **18** and the insulating layer **20'** cause the liner layer **18** and the insulating layer **20'** having different etch rates to the etchants of a subsequent recessing process, leading to potential dishing effect in the insulating layer **20'** in a subsequent etch back process.

(39) Referring to FIG. 5, a first recessing process (e.g., etch back process) is performed to remove a first portion of the liner layer **18** and a first portion of the insulating layer **20'**. The first recessing process is stopped when a bottom of the topmost second semiconductor layer **204D** is exposed. As

discussed previously with regard to FIG. 2, the topmost layer of the second nanostructures **204** (i.e., the second nanostructure **204D**) can function as a dummy semiconductor layer or sacrificial layer to protect the channel regions of GAA-FETs from being damaged during a subsequent ion implantation process and has a thickness t_3 different from the thicknesses t_2 of the other second semiconductor layers **204A-204C**, for example, greater than the thickness t_2 of a bottommost layer of the second nanostructures **204** (i.e., the second nanostructure **204A**). After etching the insulating layer **20'**, the insulating layer **20'** remains covering a topmost layer of the first nanostructures **202** (i.e., the topmost first semiconductor layer **202D**).

(40) The first recessing process may result in the dishing effect in the insulating layer **20'**. For example, the insulating layer **20'** has a concave top surface. The dishing effect may cause a recessed depth R_1 in a top surface of the insulating layer **20'**. The recessed depth R_1 is height difference between a lowest position of the top surface of the insulating layer **20'** and a highest position of the top surface of the insulating layer **20'**, wherein the lowest position is in the vicinity of a central region of the insulating layer **20'**, and the highest position is in the vicinity of the boundary between the insulating layer **20'** and the liner layer **18**. For example, the recessed depth R_1 is from 0.5 nm to 1.5 nm. In other words, the first recessing process is performed such that the insulating layer **20** has a top surface recessed into the insulating layer **20'** having a maximum depth R_1 in a range from 0.5 nm to 1.5 nm.

(41) Referring to FIG. 6A, an ion implantation process is performed to introduce impurities **22** such as Si, He and/or N atoms into the insulating layer **20'**. In some embodiments, the ion implantation process is performed to introduce the impurities **22** into an upper portion **20U** of the insulating layer **20'** without introducing impurities **22** into a lower portion **20L** of the insulating layer **20'**. The ion implantation process has an implant depth in a range from 30 nm to 50 nm. The ion implantation is used to reduce an etch rate of the insulating layer **20'**. For example, by using the ion implantation process, the insulating layer **20'** can have additional atoms (e.g., the impurities **22**) therein and thus can have an increased density. Therefore, the insulating layer **20'** can be more robust against the etchant in a subsequent second recessing process (see FIGS. 7A-7B). In some embodiments, the insulating layer **20'** may be made of silicon oxide, silicon nitride, silicon oxynitride or a combination thereof, and may include Si—O bonding, Si—N bonding, Si—Si bonding or a combination thereof. In some embodiments where the ions include N, the ion implantation process allows formation of Si—N bonding such that a total amount of the Si—N bonding in the insulating layer **20** can be increased. In some embodiments where the ions include Si, the ion implantation process allows formation of Si—Si bonding such that a total amount of the Si—Si bonding can be increased. The Si—O bonding in the insulating layer **20** can be broken by the ions during the ion implantation process. It is noted that the Si—N bonding and the Si—Si bonding have etch rates less than an etch rate of the Si—O bonding. By increasing amounts of the Si—Si bonding and the Si—N bonding, the insulating layer **20** can have reduced etch rate.

(42) In some embodiments, ion implantation process may result in the insulating layer **20'** having an impurity concentration gradient. In other words, a gradient in impurity concentration may be generated in the insulating layer **20**. For example, the insulating layer **20** have an impurity concentration decreasing in a direction toward the substrate **100**. FIG. 6B is a graph illustrating an example impurity concentration in the insulating layer **20'** as a function of a distance from a top of the insulating layer **20'**. As illustrated in FIG. 6B, the ion implantation process performed to the insulating layer **20'** may create a Gaussian distribution or a complementary error function distribution of impurities (e.g., Si, N, He) in the insulating layer **20'**.

(43) FIG. 8 is a bar chart illustrating etch rates over the liner layer **18** and the non-ion implanted/ion-implanted insulating layers **20** in various embodiments constructed according to various aspects of the present disclosure. As shown in FIG. 8, it is identified that the non-ion-implanted insulating layer **20'** and the liner layer **18** have a first etch rate difference $\Delta_{\text{sub.X1}}$ and the ion-implanted insulating layer **20'** and the liner layer **18** have a second etch rate difference

$\Delta_{\text{sub.X2}}$ less than the first etch rate difference $\Delta_{\text{sub.X1}}$.

(44) FIGS. 9-11 illustrate diagrams of penetration depths versus energy performed during the ion implantation. As discussed previously, the insulating layer **20'** may include silicon oxide, silicon nitride, silicon oxynitride or a combination thereof. In FIGS. 9-11, penetration depths of Si atoms, He atoms and N atoms in silicon oxide and silicon nitride are illustrated, respectively. Referring back to FIG. 6A, in some embodiments where the mask layer **14** is made of silicon nitride, the impurities **22**, such as Si, He and/or N, may be introduced into the mask layer **14**. Reference is made to FIGS. 6A and 9-11. Since penetration depths of the Si atoms, He atoms and N atoms in the silicon nitride are smaller than penetration depths in the silicon oxide, the penetration depth of the impurities **22** in the mask layer **14** may be smaller than a penetration depth in the insulating layer **20'** made of silicon oxide.

(45) In some embodiments where the impurities include Si atoms, the ion implantation process may be performed at an energy in a range from 15 eV to 35 eV to control the penetration depth of the Si atoms in a range from 30 nm to 50 nm. In some embodiments where the impurities include He atoms, the ion implantation process may be performed at an energy in a range from 2.5 eV to 4.5 eV to control the penetration depth of the Si atoms in a range from 30 nm to 50 nm. In some embodiments where the impurities include N atoms, the ion implantation process may be performed at an energy in a range from 10 eV to 16 eV to control the penetration depth of the Si atoms in a range from 30 nm to 50 nm. The ion implantation process may further include implanting dopants at a dose in a range from about 2×10^{14} atoms/cm² to about 2×10^{15} atoms/cm², according to another embodiment. The ion implantation process may be carried out at a temperature in a range from room temperature to 450° C.

(46) Referring back to FIG. 7A, a second recessing process (e.g., etch back process) is performed to remove a second portion of the liner layer **18** and a second portion of the insulating layer **20'** to form the STI regions **20** adjacent the fin structures **206**. The second recessing process is performed after the ion implantation process. Due to the reduced etch rate of the insulating layer **20'**, the dishing effect of the STI regions **20** is mitigated. In FIG. 7A, the dishing effect may cause a recessed depth R3 in a top surface of the insulating layer **20'**. For example, the depth R3 is less than 5 nm, such as in a range from 1 nm to 4 nm. In other words, the STI regions **20** have a top surface recessed into the STI regions **20** with the maximum depth R3, which is less than 5 nm, such as in a range from 1 nm to 4 nm. The second recessing process is performed until the bottommost first semiconductor layer **202A** is exposed. In some embodiments, the second recessing process uses a same etchant as the first recessing process, but takes a longer etch duration than the first recessing process. In FIG. 7A, the upper portion **20U** of the insulating layer **20'**, which comprises the impurities **22**, are totally removed after the second recessing process and only the lower portion **20L** of the insulating layer **20'**, which does not comprise the impurities **22** are left. In an alternative embodiment, the upper portion **20U** of the insulating layer **20'** may partially remain over the lower portion **20L** of the insulating layer **20'** after the second recessing process, as shown in FIG. 7B. The upper portion **20U** has impurities **22** while the lower portion **20L** does not have impurities **22**. Therefore, the resulting STI regions **20** have an impurity concentration gradient.

(47) FIG. 12 shows a diagram of depth R3 versus the etch rate difference $\Delta_{\text{sub.X}}$ between the liner layer **18** and the insulating layer **20'** in accordance with some embodiments. As shown in FIG. 12, there is positive correlation between the depth R3 and the etch rate difference $\Delta_{\text{sub.X}}$. By reducing the etch rate difference $\Delta_{\text{sub.X}}$ therebetween, the depth R3 of dishing can be reduced.

(48) Further in FIG. 7A, appropriate wells (not separately illustrated) may be formed in the fin structures **206**, the nanostructures **203**, and/or the STI regions **20**. In some embodiments with different well types in different device regions **1001** and **1002**, different implant steps for the first device region **1001** and the second device region **1002** may be achieved using a photoresist or other masks (not separately illustrated). For example, a photoresist may be formed over the fin structures **206** and the STI regions **20** in the first device region **1001** and the second device region **1002**. The

photoresist is patterned to expose the second device region **1002**. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, a first impurity (e.g., n-type impurity such as phosphorus, arsenic, antimony, or the like) implant is performed in the second device region **1002**, and the photoresist may act as a mask to substantially prevent the first impurities from being implanted into the first device region **1001**. After the implant, the photoresist is removed, such as by an acceptable ashing process.

(49) Following or prior to the implanting of the second device region **1002**, a photoresist or other masks (not separately illustrated) is formed over the fin structures **206**, the nanostructures **203**, and the STI regions **20** in the first device region **1001** and the second device region **1002**. The photoresist is then patterned to expose the first device region **1001**. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, a second impurity (e.g., p-type impurity such as boron, boron fluoride, indium, or the like) implant may be performed in the first device region **1001**, and the photoresist may act as a mask to substantially prevent p-type impurities from being implanted into the second device region **1002**. After the implant, the photoresist may be removed, such as by an acceptable ashing process.

(50) After one or more well implants of the first device region **1001** and the second device region **1002**, an anneal may be performed to repair implant damage and to activate the p-type and/or n-type impurities that were implanted. In some embodiments, the grown materials of epitaxial fins may be in situ doped during growth, which may obviate the implantations, although in situ and implantation doping may be used together.

(51) After the second recessing process, the mask layer **14** and the pad layer **12** are removed by using suitable etching process(s). A dummy gate dielectric layer **210** is formed on the fin structures **206** and/or the nanostructures **203**, as shown in FIG. **13**. The dummy dielectric layer **210** may be, for example, silicon oxide, silicon nitride, a combination thereof, or the like, and may be deposited or thermally grown according to acceptable techniques. A dummy gate layer **212** is formed over the dummy dielectric layer **210**, and a mask layer **214** is formed over the dummy gate layer **212**. The dummy gate layer **212** may be deposited over the dummy dielectric layer **210** and then planarized, such as by a CMP. The mask layer **214** may be deposited over the dummy gate layer **212**. The dummy gate layer **212** may be a conductive or non-conductive material and may be selected from a group including amorphous silicon, polycrystalline-silicon (polysilicon), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, and metals. The dummy gate layer **212** may be deposited PVD, CVD, sputter deposition, or other techniques for depositing the selected material. The dummy gate layer **212** may be made of other materials that have a high etching selectivity from the etching of isolation regions. The mask layer **214** may include, for example, silicon nitride, silicon oxynitride, or the like. In this example, a single dummy gate layer **212** and a single mask layer **214** are formed across the first device region **1001** and the second device region **1002**. It is noted that the dummy dielectric layer **210** is shown covering only the fin structures **206** and the nanostructures **203** for illustrative purposes only. In some embodiments, the dummy dielectric layer **210** may be deposited such that the dummy dielectric layer **210** covers the STI regions **20** and the liner layer **18**, such that the dummy dielectric layer **210** extends between the dummy gate layer **212** and the STI regions **20**.

(52) FIGS. **14A** through **25B** illustrate various following steps in the manufacturing of embodiment devices. FIGS. **15A**, **16A**, **17A**, **18A**, **19A**, **20A**, **20C**, **21A** and **21C** illustrate features in either the first device regions **1001** or the second device regions **1002**. In FIGS. **14A** and **14B**, the mask layer **214** (see FIG. **13**) may be patterned using acceptable photolithography and etching techniques to form masks **218**. The pattern of the masks **218** then may be transferred to the dummy gate layer **212** and to the dummy dielectric layer **210** to form dummy gates **216** and dummy gate dielectrics **211**, respectively. The dummy gates **216** and the dummy gate dielectrics **211** are collectively

referred to as dummy gate structures **217**. The dummy gates **216** cover respective channel regions of the fin structures **206**. The pattern of the masks **218** may be used to physically separate each of the dummy gates **216** from adjacent dummy gates **216**. The dummy gates **216** may also have a lengthwise direction substantially perpendicular to the lengthwise direction of respective fin structures **206**.

(53) In FIGS. **15A** and **15B**, a first spacer layer **220** and a second spacer layer **222** are formed over the structures illustrated in FIGS. **14A** and **14B**, respectively. The first spacer layer **220** and the second spacer layer **222** will be subsequently patterned to act as spacers for forming self-aligned source/drain regions. In FIGS. **15A** and **15B**, the first spacer layer **220** is formed on top surfaces of the STI regions **20** and the liner layer **18**; top surfaces and sidewalls of the nanostructures **203**, and the masks **218**; and sidewalls of the dummy gates **216** and the dummy gate dielectric **211**. The second spacer layer **222** is deposited over the first spacer layer **220**. The first spacer layer **220** may be formed of silicon oxide, silicon nitride, silicon oxynitride, or the like, using techniques such as thermal oxidation or deposited by CVD, ALD, or the like. The second spacer layer **222** may be formed of a material having a different etch rate than the material of the first spacer layer **220**, such as silicon oxide, silicon nitride, silicon oxynitride, or the like, and may be deposited by CVD, ALD, or the like.

(54) In FIGS. **16A** and **16B**, the first spacer layer **220** and the second spacer layer **222** are etched to form first spacers **221** and second spacers **223**. As will be discussed in greater detail below, the first spacers **221** and the second spacers **223** act to self-align subsequently formed source drain regions, as well as to protect sidewalls of the fin structures **206** and/or nanostructure **203** during subsequent processing. The first spacer layer **220** and the second spacer layer **222** may be etched using a suitable etching process, such as an isotropic etching process (e.g., a wet etching process), an anisotropic etching process (e.g., a dry etching process), or the like. In some embodiments, the material of the second spacer layer **222** has a different etch rate than the material of the first spacer layer **220**, such that the first spacer layer **220** may act as an etch stop layer when patterning the second spacer layer **222** and such that the second spacer layer **222** may act as a mask when patterning the first spacer layer **220**. For example, the second spacer layer **222** may be etched using an anisotropic etch process wherein the first spacer layer **220** acts as an etch stop layer, wherein remaining portions of the second spacer layer **222** form second spacers **223** as illustrated in FIG. **16A**. Thereafter, the second spacers **223** acts as a mask while etching exposed portions of the first spacer layer **220**, thereby forming first spacers **221** as illustrated in FIG. **16A**.

(55) As illustrated in FIG. **16A**, the first spacers **221** and the second spacers **223** are disposed on sidewalls of the nanostructures **203**. As illustrated in FIG. **16B**, in some embodiments, the second spacer layer **222** may be removed from over the first spacer layer **220** adjacent the masks **218**, the dummy gates **216**, and the dummy gate dielectrics **211**, and the first spacers **221** are disposed on sidewalls of the masks **218**, the dummy gates **216**, and the dummy gate dielectrics **211**. In other embodiments, a portion of the second spacer layer **222** may remain over the first spacer layer **220** adjacent the masks **218**, the dummy gates **216**, and the dummy gate dielectrics **211**.

(56) The above disclosure generally describes a process of forming spacers. Other processes and sequences may be used. For example, fewer or additional spacers may be utilized, different sequence of steps may be utilized (e.g., the first spacers **221** may be patterned prior to depositing the second spacer layer **222**), additional spacers may be formed and removed, and/or the like. Furthermore, devices in first device region **1001** and devices in the second device region **1002** may be formed using different structures and steps.

(57) In FIGS. **17A** and **17B**, source/drain recesses **226** are formed in the nanostructures **203** and the substrate **100**, in accordance with some embodiments. Epitaxial source/drain regions will be subsequently formed in the source/drain recesses **226**. The source/drain recesses **226** may extend through the first nanostructures **202** and the second nanostructures **204**, and into the substrate **100**. As illustrated in FIG. **17A**, bottom surfaces of the source/drain recesses **226** may be substantially

level with top surfaces of the STI regions **20**, as an example. In some other embodiments, the fin structures **206** may be etched such that bottom surfaces of the source/drain recesses **226** are disposed below the top surfaces of the STI regions **20**, or above the top surfaces of the STI regions **20**. The source/drain recesses **226** may be formed by etching the nanostructures **203** and the substrate **100** using anisotropic etching processes, such as RIE, NBE, or the like. The first spacers **221**, the second spacers **223**, and the masks **218** mask portions of the fin structures **206**, the nanostructures **203**, and the substrate **100** during the etching processes used to form the source/drain recesses **226**. A single etch process or multiple etch processes may be used to etch each layer of the nanostructures **203** and/or the fin structures **206**. Timed etch processes may be used to stop the etching of the source/drain recesses **226** after the source/drain recesses **226** reach a target depth.

(58) In FIGS. **18A** and **18B**, portions of sidewalls of the layers of the nanostructures **203** formed of the first semiconductor materials (e.g., the first nanostructures **202**) exposed by the source/drain recesses **226** are etched to form sidewall recesses **228** between corresponding second nanostructures **204**. Although sidewalls of the first nanostructures **202** in recesses **228** are illustrated as being straight in FIG. **18B**, the sidewalls may be concave or convex. The sidewalls may be etched using isotropic etching processes, such as wet etching or the like. In some embodiments in which the first nanostructures **202** include, e.g., SiGe, and the second nanostructures **204** include, e.g., Si or SiC, a dry etch process with tetramethylammonium hydroxide (TMAH), ammonium hydroxide (NH₄OH), or the like may be used to etch sidewalls of the first nanostructures **202**.

(59) In FIGS. **19A** and **19B**, inner spacers **230** are formed in the sidewall recess **228**. The inner spacers **230** may be formed by depositing an inner spacer layer (not separately illustrated) over the structures illustrated in FIGS. **18A** and **18B**. The inner spacers **230** act as isolation features between subsequently formed source/drain regions and gate structure. As will be discussed in greater detail below, source/drain regions will be formed in the recesses **226**, and the first nanostructures **202** will be replaced with corresponding gate structures.

(60) The inner spacer layer may be deposited by a conformal deposition process, such as CVD, ALD, or the like. The inner spacer layer may comprise a material such as silicon nitride or silicon oxynitride, although any suitable material, such as low-dielectric constant (low-k) materials having a k-value less than about 3.5, may be utilized. The inner spacer layer may then be anisotropically etched to form the inner spacers **230**. Although outer sidewalls of the inner spacers **230** are illustrated as being flush with sidewalls of the second nanostructures **204**, the outer sidewalls of the inner spacers **230** may extend beyond or be recessed from sidewalls of the second nanostructures **204**.

(61) Moreover, although the outer sidewalls of the inner spacers **230** are illustrated as being straight in FIG. **19B**, the outer sidewalls of the inner spacers **230** may be concave or convex. The inner spacer layer may be etched by an anisotropic etching process, such as RIE, NBE, or the like. The inner spacers **230** may be used to prevent damage to subsequently formed source/drain regions (such as the epitaxial source/drain regions **232**, discussed below with respect to FIGS. **20A** and **20B**) by subsequent etching processes, such as etching processes used to form gate structures.

(62) Epitaxial source/drain regions **232** are formed in the source/drain recesses **226**, as shown in FIGS. **20A** and **20B**. In some embodiments, the source/drain regions **232** may exert stress on the second nanostructures **204**, thereby improving device performance. As illustrated in FIG. **20B**, the epitaxial source/drain regions **232** are formed in the source/drain recesses **226** such that each dummy gate **216** is disposed between respective neighboring pairs of the epitaxial source/drain regions **232**. In some embodiments, the first spacers **221** are used to separate the epitaxial source/drain regions **232** from the dummy gate layer **212** and the inner spacers **230** are used to separate the epitaxial source/drain regions **232** from the first nanostructures **202** by an appropriate lateral distance so that the epitaxial source/drain regions **232** do not short out with subsequently

formed gates of the resulting GAA-FETs.

(63) In some embodiments, the epitaxial source/drain regions **232** may include any acceptable material appropriate for n-type GAA-FETs. For example, if the second nanostructures **204** are silicon, the epitaxial source/drain regions **232** may include materials exerting a tensile strain on the second nanostructures **204**, such as silicon carbide, phosphorous doped silicon carbide, silicon phosphide, or the like. In some embodiments, the epitaxial source/drain regions **232** may include any acceptable material appropriate for p-type GAA-FETs. For example, if the second nanostructures **204** are silicon, the epitaxial source/drain regions **232** may comprise materials exerting a compressive strain on the second nanostructures **204**, such as silicon germanium, boron doped silicon germanium, germanium, germanium tin, or the like. The epitaxial source/drain regions **232** may have surfaces raised from respective upper surfaces of the nanostructures **203** and may have facets.

(64) The epitaxial source/drain regions **232** may be implanted with dopants to form source/drain regions, followed by an anneal. The n-type and/or p-type impurities for source/drain regions may be any of the impurities previously discussed. In some embodiments, the epitaxial source/drain regions **232** may be in situ doped during growth.

(65) As a result of the epitaxy processes used to form the epitaxial source/drain regions **232**, upper surfaces of the epitaxial source/drain regions **232** have facets which expand laterally outward beyond sidewalls of the nanostructures **203**. In some embodiments, these facets cause adjacent epitaxial source/drain regions **232** to merge as illustrated by FIG. **20A**. In some other embodiments, adjacent epitaxial source/drain regions **232** remain separated after the epitaxy process is completed as illustrated by FIG. **20C**. In the embodiments illustrated in FIGS. **20A** and **20C**, the first spacers **221** may be formed to a top surface of the STI regions **20** and the liner layer **18** thereby blocking the lateral epitaxial growth. In some other embodiments, the first spacers **221** may cover portions of the sidewalls of the nanostructures **203** further blocking the epitaxial growth. In some other embodiments, the spacer etch used to form the first spacers **221** may be adjusted to remove the spacer material to allow the epitaxially grown region to extend to the surface of the STI region **20**.

(66) The epitaxial source/drain regions **232** may comprise one or more semiconductor material layers. For example, the epitaxial source/drain regions **232** may comprise a first semiconductor material layer **232A**, a second semiconductor material layer **232B**, and a third semiconductor material layer **232C**. Any number of semiconductor material layers may be used for the epitaxial source/drain regions **232**. Each of the first semiconductor material layer **232A**, the second semiconductor material layer **232B**, and the third semiconductor material layer **232C** may be formed of different semiconductor materials and may be doped to different dopant concentrations. In some embodiments, the first semiconductor material layer **232A** may have a dopant concentration less than the second semiconductor material layer **232B** and greater than the third semiconductor material layer **232C**. In embodiments in which the epitaxial source/drain regions **232** comprise three semiconductor material layers, the first semiconductor material layer **232A** may be deposited, the second semiconductor material layer **232B** may be deposited over the first semiconductor material layer **232A**, and the third semiconductor material layer **232C** may be deposited over the second semiconductor material layer **232B**.

(67) In FIGS. **21A-21C**, an interlayer dielectric (ILD) layer **236** is deposited over the structure illustrated in FIGS. **20A-20C**. The ILD layer **236** may be formed of a dielectric material, and may be deposited by any suitable method, such as CVD, plasma-enhanced CVD (PECVD), or FCVD. Dielectric materials may include phospho-silicate glass (PSG), boro-silicate glass (BSG), boron-doped phospho-silicate glass (BPSG), undoped silicate glass (USG), or the like. Other insulation materials formed by any acceptable process may be used. In some embodiments, a contact etch stop layer (CESL) **234** is disposed between the ILD layer **236** and the epitaxial source/drain regions **232**, the masks **214**, and the first spacers **221**. The CESL **234** may comprise a dielectric material, such as, silicon nitride, silicon oxide, silicon oxynitride, or the like, having a different etch rate than

the material of the overlying ILD layer **236**.

(68) A planarization process, such as a CMP, may be performed to remove the masks **218** on the dummy gates **216**, the first spacers **221** along sidewalls of the masks **218** and the dummy gate dielectrics **211**. The planarization process levels the top surface of the ILD layer **236** with the top surfaces of the topmost second semiconductor layer **204D**. The planarization process may further go on to remove top portions of the topmost second semiconductor layer **204D**, the CESL **234** and the ILD layer **236**, as shown in FIGS. **22A** and **22B**, to reach a desired thickness of a subsequently formed gate electrode (see FIG. **25B**). After the planarization process, top surfaces of the topmost second semiconductor layer **204D**, and the ILD layer **236** are level within process variations. Accordingly, the top surfaces of the topmost second semiconductor layer **204D** are exposed through the ILD layer **236**.

(69) In FIGS. **23A** and **23B**, the topmost second semiconductor layer **204D** are removed in one or more etching steps, so that gate trenches **238** are formed between corresponding CESL **234**. In some embodiments, the topmost second semiconductor layers **204D** are removed by an anisotropic dry etch process. For example, the etching process may include a dry etch process using reaction gas(es) that selectively etch the topmost second semiconductor layer **204D** at a faster rate than the ILD layer **236**. Each gate trench **238** exposes and/or overlies portions of nanostructures **203'**, which act as channel regions in subsequently completed GAA-FETs. The nanostructures **203'** which act as the channel regions are disposed between neighboring pairs of the epitaxial source/drain regions **232**. During the removal, the topmost first semiconductor layer **202D** may be used as etch stop layers when the topmost second semiconductor layer **204D** are etched.

(70) In FIG. **23C**, sidewall spacers **250** may be disposed on sidewalls of the CESL **234**. In some embodiments, a spacer layer is formed over the structure illustrated in FIGS. **22A** and **22B** and subsequently etched to form the sidewall spacers **250**. The spacer layer may be formed of silicon oxide, silicon nitride, silicon oxynitride, or the like, using techniques such as thermal oxidation or deposited by CVD, ALD, or the like. The spacer layer may then be etched using an anisotropic etch process in which remaining portions of the spacer layer form sidewall spacers **250**.

(71) In FIGS. **24A** and **24B**, the first nanostructures **202** in the gate trenches **238** are removed by an isotropic etching process such as wet etching or the like using etchants which are selective to the materials of the first nanostructures **202**. Stated differently, the first nanostructures **202** are removed by using a selective etching process that etches the first nanostructures **202** at a faster etch rate than it etches the second nanostructures **204**, thus forming spaces between the second nanostructures **204** (also referred to as sheet-sheet spaces if the nanostructures **204** are nanosheets). This step can be referred to as a channel release process. As illustrated in FIGS. **24A** and **24B**, gaps **239** (empty spaces) are formed between the second nanostructures **204**. At this interim processing step, the gaps **239** between second nanostructures **204** may be filled with ambient environment conditions (e.g., air, nitrogen, etc). In some embodiments, the second nanostructures **204** can be referred to as nanosheets, nanowires, nanoslabs, nanorings having nano-scale size (e.g., a few nanometers), depending on their geometry. For example, in some embodiments the second nanostructures **204** may be trimmed to have a substantial rounded shape (i.e., cylindrical) due to the selective etching process for completely removing the first nanostructures **202**. In that case, the resultant second nanostructures **204** can be called nanowires.

(72) In embodiments in which the first nanostructures **202** include, e.g., SiGe, and the second nanostructures **204** include, e.g., Si or SiC, tetramethylammonium hydroxide (TMAH), ammonium hydroxide (NH₄OH) or the like may be used to remove the first nanostructures **202**. In some embodiments, both the channel release step and the previous step of laterally recessing first nanostructures **202** (i.e., the step as illustrated in FIGS. **18A-18B**) use a selective etching process that etches first nanostructures **202** (e.g., SiGe) at a faster etch rate than etching second nanostructures **204** (e.g., Si), and therefore these two steps may use the same etchant chemistry in some embodiments. In this case, the etching time/duration of channel release step is longer than the

etching time/duration of the previous step of laterally recessing first nanostructures **202**, so as to completely remove the sacrificial nanostructures **202**.

(73) Next, in FIGS. **25A** and **25B**, high-k/metal gate structures are formed. For example, a gate dielectric layer **240** is formed (e.g., conformally) in the gate trenches **238** and in the gaps **239**. The gate dielectric layer **240** wraps around the second nanostructures **204**, lines sidewalls of the inner spacers **230** and sidewalls of the first spacers **211**, and extends along the upper surface of the fin structures **206**. In accordance with some embodiments, the gate dielectric layer **120** comprises silicon oxide, silicon nitride, or multilayers thereof. In some embodiments, the gate dielectric layer **240** includes a high-k dielectric material, and in these embodiments, the gate dielectric layer **240** may have a dielectric constant greater than about 7.0, and may include a metal oxide or a silicate of Hf, Al, Zr, La, Mg, Ba, Ti, or Pb, or combinations thereof. The formation methods of the gate dielectric layer **240** may include Molecular-Beam Deposition (MBD), ALD, PECVD, and the like.

(74) In an alternative embodiment, an interfacial layer (not shown) is deposited between the gate dielectric layer **240** and the second nanostructures **204** and is formed of silicon oxide or silicon oxynitride grown by a thermal oxidation process. For example, the interfacial layer can be grown by a rapid thermal oxidation (RTO) process or by an annealing process using oxygen.

(75) Next, a gate electrode material (e.g., an electrically conductive material) is formed in the gate trenches **238** and in the gaps **239** to form the gate electrodes **242**. The gate electrodes **242** fill the remaining portions of the gate trenches **238** and in the gaps **239**. For example, the gate electrodes **242** include one or more work function layers **244** and a fill metal layer **246**. A CMP is then performed on the fill metal layer **246**, the one or more work function layers **244** and the gate dielectric layer **240** until the ILD layer **236** is exposed, resulting in the fill metal layer **246**, the one or more work function layers **244** and the gate dielectric layer **240**, the CESL **234**, and the ILD layer **236** having substantially level top surfaces. The gate electrodes **242** and the gate dielectric layer **240** are collectively referred to as metal gate structures **248**.

(76) The one or more work function layers **244** may be deposited to surround each of the second nanostructures **204**. A portion of the one or more work function layers **244** is formed vertically between adjacent second nanostructures **204** and fills the gaps **239** between adjacent second nanostructures **204**.

(77) The one or more work function layers can provide a suitable work function for the high-k/metal gate structures. For an n-type GAA FET, the one or more work function layers **244** may include one or more n-type work function metals (N-metal). The n-type work function metals may exemplarily include, but are not limited to, titanium aluminide (TiAl), titanium aluminium nitride (TiAlN), carbo-nitride tantalum (TaCN), hafnium (Hf), zirconium (Zr), titanium (Ti), tantalum (Ta), aluminum (Al), metal carbides (e.g., hafnium carbide (HfC), zirconium carbide (ZrC), titanium carbide (TiC), aluminum carbide (AlC)), aluminides, and/or other suitable materials. On the other hand, for a p-type GAA FET, the one or more work function layers **244** may include one or more p-type work function metals (P-metal). The p-type work function metals may exemplarily include, but are not limited to, titanium nitride (TiN), tungsten nitride (WN), tungsten (W), ruthenium (Ru), palladium (Pd), platinum (Pt), cobalt (Co), nickel (Ni), conductive metal oxides, and/or other suitable materials.

(78) In some embodiments, the fill metal layer **246** may exemplarily include, but are not limited to, tungsten, aluminum, copper, nickel, cobalt, titanium, tantalum, titanium nitride, tantalum nitride, nickel silicide, cobalt silicide, TaC, TaSiN, TaCN, TiAl, TiAlN, or other suitable materials.

(79) Based on the above discussions, it can be seen that the present disclosure in various embodiments offers advantages. It is understood, however, that other embodiments may offer additional advantages, and not all advantages are necessarily disclosed herein, and that no particular advantage is required for all embodiments. One advantage is that by using the ion implantation process, the insulating layer for forming the shallow trench isolation region can have reduced etch rate. Another advantage is that the reduced etch rate can aid in reducing an etch rate

difference between the liner layer and the insulating layer. Another advantage is the reduced etch rate difference may be advantageous for mitigating the dishing effect of the STI region.

(80) In some embodiments, a method of forming a semiconductor device includes etching trenches in a substrate to form fin structures, depositing a liner layer to line the trenches, filling the trenches with an insulating layer, performing an ion implantation process to the insulating layer, after performing the ion implantation process, recessing the insulating layer to form shallow trench isolation (STI) regions adjacent the fin structures, and forming a gate crossing the fin structures. In some embodiments, the method further comprises etching the insulating layer prior to the ion implantation process. In some embodiments, the method further comprises forming a multi-layer stack on the substrate, the multi-layer stack comprising alternately stacking first semiconductor layers and second semiconductor layers, and patterning the multi-layer stack to form first nanostructures and second nanostructures, wherein after etching the insulating layer, the insulating layer remains covering a topmost layer of the first nanostructures. In some embodiments, after recessing the insulating layer, the STI regions have a top surface recessed into the STI regions with a maximum depth of less than 5 nm. In some embodiments, performing the ion implantation process comprises: introducing impurities comprising Si atoms, He atoms, N atoms or a combination thereof into the insulating layer. In some embodiments, an implant depth of the ion implantation process is in a range from 30 nm to 50 nm. In some embodiments, the ion implantation process is performed to introduce impurities into an upper portion of the insulating layer without introducing the impurities into a lower portion of the insulating layer. In some embodiments, recessing the insulating layer is performed to remove the upper portion without removing the lower portion.

(81) In some embodiments, a method of forming a semiconductor device includes forming a multi-layer stack on a substrate, wherein the multi-layer stack comprises alternately stacked first semiconductor layers and second semiconductor layers, etching trenches in the multi-layer stack and the substrate to form fins each comprising alternately stacked first nanostructures and second nanostructures, filling the trenches with an insulating layer, performing a first recessing process to the insulating layer such that a topmost one of the second nanostructures is exposed, implanting the insulating layer using ions, performing a second recessing process to the insulating layer to form shallow trench isolation (STI) regions below the first and second nanostructures, and forming a gate structure in place of the first nanostructures. In some embodiments, the topmost one of the second nanostructures has a thickness greater than a thickness of a topmost one of the first nanostructures. In some embodiments, the topmost one of the second nanostructures has a thickness greater than a thickness of another one of the second nanostructures. In some embodiments, the method further comprises prior to etching the trenches in the multi-layer stack and the substrate, forming a pad layer and a mask layer in sequence over the multi-layer stack, and during the first recessing process, the pad layer and the mask layer remain over the multi-layer stack. In some embodiments, during implanting the insulating layer, the pad layer and the mask layer remain over the multi-layer stack. In some embodiments, during the second recessing process, the pad layer and the mask layer remain over the multi-layer stack. In some embodiments, the pad layer and the mask layer are removed after the second recessing process. In some embodiments, the first recessing process is stopped when a bottom of the topmost one of the second nanostructures is exposed. In some embodiments, the first recessing process is performed such that the insulating layer has a top surface recessed into the insulating layer having a maximum depth in a range from 0.5 nm to 1.5 nm. In some embodiments, the topmost one of the second nanostructures is removed before forming the gate structure.

(82) In some embodiments, a semiconductor device includes a fin structure protruding from a substrate, a shallow trench isolation (STI) region adjacent the fin structure, a gate structure crossing the fin structure and source/drain regions on opposite sides of the gate structure. The STI region has impurities having an impurity concentration gradient. In some embodiments, the impurities

comprise Si, N, He or a combination thereof.

(83) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method of forming a semiconductor device, comprising: etching trenches in a substrate to form fin structures; depositing a liner layer to line the trenches; filling the trenches with an insulating layer; performing an ion implantation process to the insulating layer; after performing the ion implantation process, recessing the insulating layer to form shallow trench isolation (STI) regions adjacent the fin structures; and forming a gate crossing the fin structures.
2. The method of claim 1, further comprising: etching the insulating layer prior to the ion implantation process.
3. The method of claim 2, further comprising: forming a multi-layer stack on the substrate, the multi-layer stack comprising alternately stacking first semiconductor layers and second semiconductor layers; and patterning the multi-layer stack to form first nanostructures and second nanostructures, wherein after etching the insulating layer, the insulating layer remains covering a topmost layer of the first nanostructures.
4. The method of claim 1, wherein after recessing the insulating layer, the STI regions have a top surface recessed into the STI regions with a maximum depth of less than 5 nm.
5. The method of claim 1, wherein performing the ion implantation process comprises: introducing impurities comprising Si atoms, He atoms, N atoms or a combination thereof into the insulating layer.
6. The method of claim 1, wherein an implant depth of the ion implantation process is in a range from 30 nm to 50 nm.
7. The method of claim 1, wherein the ion implantation process is performed to introduce impurities into an upper portion of the insulating layer without introducing the impurities into a lower portion of the insulating layer.
8. The method of claim 7, wherein recessing the insulating layer is performed to remove the upper portion without removing the lower portion.
9. A method of forming a semiconductor device, comprising: forming a multi-layer stack on a substrate, wherein the multi-layer stack comprises alternately stacked first semiconductor layers and second semiconductor layers; etching trenches in the multi-layer stack and the substrate to form fins each comprising alternately stacked first nanostructures and second nanostructures; filling the trenches with an insulating layer; performing a first recessing process to the insulating layer such that a topmost one of the second nanostructures is exposed; implanting the insulating layer using ions; performing a second recessing process to the insulating layer to form shallow trench isolation (STI) regions below the first and second nanostructures; and forming a gate structure in place of the first nanostructures.
10. The method of claim 9, wherein the topmost one of the second nanostructures has a thickness greater than a thickness of a topmost one of the first nanostructures.
11. The method of claim 9, wherein the topmost one of the second nanostructures has a thickness greater than a thickness of another one of the second nanostructures.
12. The method of claim 9, further comprising: prior to etching the trenches in the multi-layer stack

and the substrate, forming a pad layer and a mask layer in sequence over the multi-layer stack; and during the first recessing process, the pad layer and the mask layer remain over the multi-layer stack.

13. The method of claim 12, wherein during implanting the insulating layer, the pad layer and the mask layer remain over the multi-layer stack.

14. The method of claim 12, wherein during the second recessing process, the pad layer and the mask layer remain over the multi-layer stack.

15. The method of claim 12, wherein the pad layer and the mask layer are removed after the second recessing process.

16. The method of claim 9, wherein the first recessing process is stopped when a bottom of the topmost one of the second nanostructures is exposed.

17. The method of claim 9, wherein the first recessing process is performed such that the insulating layer has a top surface recessed into the insulating layer having a maximum depth in a range from 0.5 nm to 1.5 nm.

18. The method of claim 9, wherein the topmost one of the second nanostructures is removed before forming the gate structure.

19. A method of forming a semiconductor device, comprising: forming a fin structure protruding from a substrate; forming a shallow trench isolation (STI) region adjacent the fin structure, wherein the STI region has impurities having an impurity concentration gradient; forming a gate structure crossing the fin structure; and forming source/drain regions on opposite sides of the gate structure.

20. The method of claim 19, wherein the impurities comprise Si, N, He or a combination thereof.
